PART 1: ADAPTIVE SYSTEM ARCHITECTURE & SECURITY BOUNDARY

SYSTEM BLUEPRINT

A transparent, deterministic knowledge-tracing architecture engineered for Indian school classrooms (40–60 students per teacher). Strict physical and logical decoupling between the deterministic evaluation engine and the Socratic AI tutor companion.

LAYER 1 Client & Presentation Layer (Offline-First Next.js 16 + React 19)

MINIMAL MONOCHROME UI

Student Learning Portal `/student`

Daily study streak tracking, 3-star concept ratings, 15-question diagnostic runner, visual fraction bars, and 1-click WhatsApp parent progress reports.

Teacher Copilot Dashboard `/teacher`

Real-time student mastery matrix heatmap, classroom-wide bottleneck alerts (e.g. LCM deficit), and pedagogical intervention override controls.

Offline PWA Sync Engine `IndexedDB`

Caches full NCERT lessons & questions for rural schools. Queues offline student attempts with client UUIDv4 for idempotent auto-sync upon reconnection.

LAYER 2 Edge Security, Anti-Tamper & Role-Based Authorization

ZERO-TRUST BOUNDARY

Middleware Session Guard

Enforces HTTP-only SameSite cookies via `@supabase/ssr`. Blocks unverified teachers and routes them to `/pending-approval`.

Zero Client Grading

Client never submits scores or passes percentages. Client submits only selected option IDs; grading occurs strictly on the protected server runtime.

Production Security Headers

Strict CSP without unsafe external domains, HSTS (63M seconds), `X-Frame-Options: DENY`, and anti-open-redirect destination validators.

LAYER 3 Deterministic Adaptive Core Engine (`lib/adaptive/`)

AUDITABLE & TRANSPARENT

1. Prerequisite DAG Graph

Acyclic Directed Graph mapping Class 7 Fractions:

- **C1:** Fraction Basics (Parts of a whole)
- **C2:** Common Multiples & Factors (LCM)
- **C3:** Equivalent Fractions (Req: C1)
- **C4:** Adding Fractions (Req: C2 + C3)
- **C5:** Mixed Fraction Problems (Req: C4)

2. Deterministic Mastery Calculation

Strict ratio formula per concept node:

```
Mastery(c) = CorrectAttempts / TotalValidEvidence
```

- **Mastered:** ≥ 80% with ≥ 2 valid attempts
- **Partial:** 50% to < 80% (or ≥ 80% with 1 attempt)
- **Needs Revision:** < 50% score

3. Root Gap Prerequisite Traversal

Recursive DFS inspects all transitive prerequisites:
If student struggles with C4 (Adding Fractions), engine inspects C2 & C3. If C2 (LCM) < 80%, engine flags **Root Prerequisite Bottleneck** and routes to C2 first!

ARCHITECTURAL FIREWALL: STRICT DECOUPLING *The LLM is NEVER the grader. Deterministic algorithms decide mastery; AI provides only Socratic hints.*

Socratic AI Companion Service (`lib/ai/`)

LLM GUARDRAILS

Server-Side Only: Zero secret leakage. Injects concept context, objective, and Indian cultural analogies (dividing rotis, sharing dosas, cricket over run-rates). Rate-limited to 15 req/min per student.

Socratic Prompt Guardrails

ZERO SPOON-FEEDING

The AI is instructed strictly to ask probing questions and guide students step-by-step. **Private answer keys are strictly withheld** from the AI context payload to completely eliminate prompt leakage.

LAYER 4 Data Persistence, Row Level Security & Real-Time Telemetry (Supabase PostgreSQL)

ACID COMPLIANCE

Row Level Security (RLS)

Students access only own attempts. Teachers view only enrolled class cohorts. Database-level policy prevents foreign profile tampering.

PostgreSQL Realtime CDC

Subscribes to `student_mastery` and `student_attempts` events, instantly updating Teacher Copilot telemetry with zero page refresh.

Immutable Audit Trail

Captures all educator approvals, account suspensions, and teacher intervention assignments with actor ID and ISO timestamps.

1. Deterministic & Explainable
Every recommendation contains a deterministic audit rationale explaining why a prerequisite was assigned (e.g. *"C4 requires C2 (LCM), currently at 45%"*).

2. Privacy of Minors
Zero commercial analytics or ad trackers. Student PII is strictly isolated; student names and emails are never forwarded to external AI providers.

3. Classroom Reality
Engineered for real Indian state board and CBSE schools: erratic internet handling, high student-to-teacher ratios, and multi-language support (EN, HI, MR).

PART 2: ADAPTIVITY WORKFLOW & DYNAMIC REMEDIATION TRAVERSAL

EXECUTION PIPELINE

Detailed step-by-step lifecycle showing how a student attempt triggers deterministic grading, DAG gap traversal, dynamic three-way personalized branching, Socratic scaffolding, and real-time teacher feedback loops.

1 Diagnostic Assessment Administration

15-QUESTION BASELINE

Student begins with a 15-question baseline assessment (3 randomized questions per concept across all 5 DAG nodes). The client receives question stems and options; **private answer keys remain strictly on the server** to prevent browser inspection cheating.

2 Server-Side Anti-Tamper Evaluation & Attempt Deduplication

API: /API/ASSESSMENT/SUBMIT

Client submits answers payload `[{ questionId, selectedOptionId, timeSpentSeconds }]`. Server verifies submissions against private `ANSWER_KEYS`, computes isCorrect, persists raw attempts to `student_attempts`, and deduplicates multiple attempts on identical questions.

3 Deterministic Node Mastery Evaluation

FORMULA EVALUATION

The mathematical engine evaluates each concept node independently: `Mastery(c) = Correct / ValidEvidence`. Threshold evaluation: MASTERED (≥80% + N≥2), PARTIAL (50%–79%), NEEDS REVISION (<50%). Persists to `student_mastery` database table.

4 Prerequisite DAG Gap Traversal Algorithm

DFS PREREQUISITE CHECK

For current target concept *C*, the engine executes a topological Depth-First Search over all upstream prerequisites:

```
for each P in TransitivePrerequisites(C): if Mastery(P) < 0.80 then FLAG_BOTTLENECK(P)
```

If multiple prerequisites are weak, the engine selects the foundational root node first (tie-breaker: lowest curriculum order index).

5 Dynamic Pedagogical Routing (Three Individualized Journeys)

DIFFERENTIATED LEARNING

Based on the DAG audit, students are partitioned into three distinct learning trajectories:

Branch A: Accelerated — 94% SCORE
Student: Aarav Gupta
All foundational nodes mastered. Bypasses repetitive procedural drills and advances straight to multi-step *Mixed Fraction Word Problems* and algebraic applications.

Branch B: Guided Scaffolding — 72% SCORE
Student: Diya Patel
Prerequisites intact, but occasional reduction errors. Assigned visual fraction bars and guided step-by-step practice to cement fraction simplification fluency.

Branch C: Root Gap Bridge — 48% SCORE
Student: Rohan Sharma
Failure in Adding Fractions is diagnosed as a *Common Multiples (LCM at 45%)* deficit. Routed to a 5-minute visual LCM module before returning to fraction addition.

6 Socratic Scaffolding & Conceptual Remediation

INTERACTIVE PRACTICE

Students practice targeted questions. When stuck, the Socratic AI Companion provides step-by-step hints and relatable Indian cultural analogies (e.g. *"Think of sharing a 6-slice pizza among friends"* or *"Finding the LCM of bell ringing intervals"*) without ever disclosing the correct answer choice.

7 Continuous Mastery Re-evaluation & Real-Time Telemetry Feedback Loop

CLOSED LOOP

Upon completing remediation, new attempt records update concept mastery. When the prerequisite gap is cleared (≥ 80%), the engine unlocks downstream concepts, broadcasts a live update to the **Teacher Copilot Matrix**, and generates an updated **Parent WhatsApp Progress Card**.

ID	Curriculum Concept	Prerequisites	Downstream Impact	Root Remediation Action
C1	Fraction Basics	None (Foundational)	Blocks C3, C4, C5	Visual fraction parts-of-a-whole interactive model
C2	Common Multiples & Factors	None (Arithmetic)	Blocks C4, C5	Common multiples list & prime factorization tree
C3	Equivalent Fractions	C1 (Fraction Basics)	Blocks C4, C5	Multiplying/dividing numerator & denominator grids
C4	Adding Fractions	C2 (LCM) + C3 (Equivalent)	Blocks C5	Convert unlike denominators to common LCM base
C5	Mixed Fraction Problems	C4 (Adding Fractions)	Curriculum Summit	Decompose mixed fractions to improper fractions